# Paid media: what it returns, and how to spend next quarter's 18M

To: CMO, CFO · From: Data Science · 27 Oct 2026 · Based on 40 regions × 156 weeks to 2024-01-15. Every figure is reproduced by one command (*make all*); amounts in the data's currency.

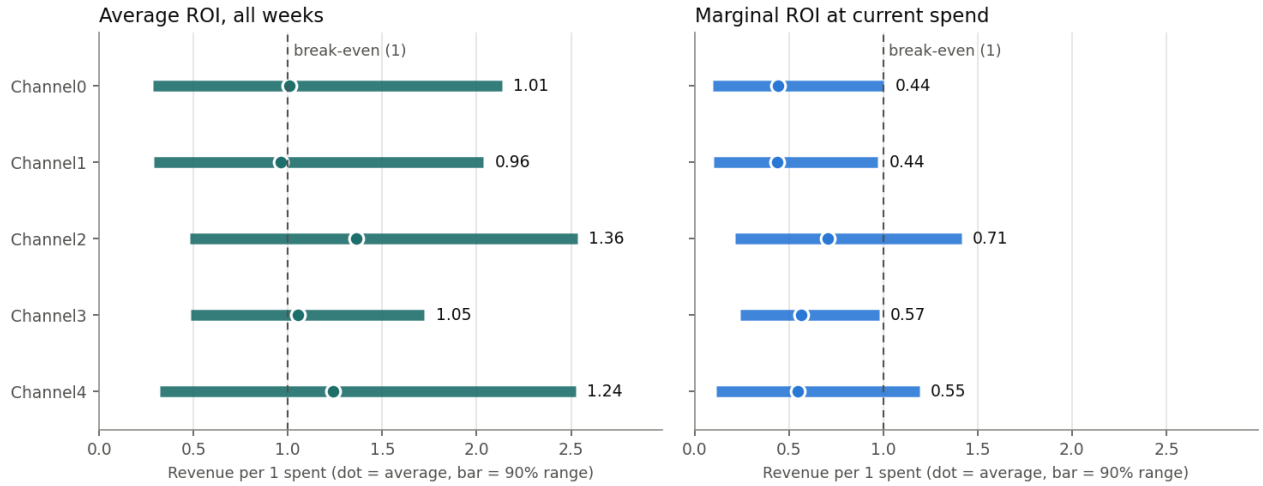
**Recommendation in one minute**

- **Make the cut to 18M.** At today's spend, the next 1 in any channel brings back only 0.44–0.71 in revenue (chance it pays back: 4%–19%). Going from 20.77M to 18M saves 2.77M of spend and gives up about 1.46M of revenue (90%: 0.74M–2.31M).
- **Tilt the cut**: Channel1 -29%, Channel0 -29%; Channel4 -10%, Channel3 -8%; Channel2 +20%. This adds about 0.11M over cutting every channel equally (66% chance it's better), and holds whatever we assume about how fast returns fall. The cut itself is the big decision; the split is a sensible tilt.
- **Do not cut Channel3 because of the vendor's 0.8.** Our own spend records show Channel3 was never switched off in the test regions (2 of 64 test weeks at zero; spend +18%). Replace it with a clean test (12 weeks, 20 regions, cost about 181k).

## 1. What each channel returns

| Channel | Spend, last 13 wks | Average return per 1 (3 years, 90% range) | Return on the *next* 1 (today's spend, 90% range) | Chance it pays back | Share of revenue |
|---|---|---|---|---|---|
| Channel0 | 3.80M | 1.01 (0.29–2.13) | 0.44 (0.10–1.00) | 5% | 3.1% |
| Channel1 | 3.20M | 0.96 (0.29–2.04) | 0.44 (0.10–0.97) | 4% | 2.3% |
| Channel2 | 1.44M | 1.36 (0.48–2.53) | 0.71 (0.22–1.41) | 19% | 1.2% |
| Channel3 | 7.95M | 1.05 (0.49–1.72) | 0.57 (0.24–0.98) | 4% | 7.0% |
| Channel4 | 4.37M | 1.24 (0.32–2.52) | 0.55 (0.11–1.19) | 11% | 4.5% |
| **All paid media** | **20.77M** | **1.09 (0.75–1.46)** | **–** | **–** | **18% (12%–24%)** |

**How to read this.** On average over three years, paid media roughly pays back its cost in revenue (1.09) and drives about 18% of revenue. But every channel is past the point where more spend pays for itself: the *next* 1 returns about half. These are **revenue** returns; to make **profit** a channel needs a return above 1 ÷ gross margin (e.g. 2.5 at a 40% margin), so the case against adding spend is stronger still. Ranges are wide and overlap, so treat channel rankings as tendencies, not verdicts.




Left: average return per 1 over three years. Right: return on the next 1 at today's spend. Dot = best estimate, bar = 90% range, dashed line = break-even in revenue.

## 2. The 18M plan (each channel within ±30% of last quarter)

| Channel | Last 13 weeks | Next quarter | Change | If returns fall fast / slowly |
|---|---|---|---|---|
| Channel0 | 3.80M | 2.70M | -29% | -29% to -28% |
| Channel1 | 3.20M | 2.27M | -29% | -29% to -29% |
| Channel2 | 1.44M | 1.74M | +20% | +14% to +25% |
| Channel3 | 7.95M | 7.35M | -8% | -9% to -7% |
| Channel4 | 4.37M | 3.94M | -10% | -13% to -4% |
| **Total** | **20.77M** | **18M** | **-13%** | |

Expected extra revenue from paid media next quarter: **20.9M** (90%: 14.4M–27.6M), versus 20.8M for an equal cut everywhere. The plan assumes next quarter is bought like the last one (same weekly timing and prices).

## 3. If the budget is 12M or 24M

- **12M is not possible within the ±30% rule**: the lowest total the rule allows is 14.54M. Choose which rule to break: cut every channel by 42%, or cut deeper where it hurts least (Channel0 and Channel1 by more than half). Either way, expect 16.5M–16.6M of extra revenue; the 6M saved vs 18M costs about 4.27M of revenue (0.71 per 1).
- **24M**: the extra 3.23M above today's spend would bring back about 1.71M (0.53 per 1; chance it pays back 0%). Not recommended on revenue grounds.

## 4. The Channel3 experiment, and the test to run instead

The vendor reported that every 1 on Channel3 returned 0.8. Our records show the switch-off didn't happen (2 of 64 test region-weeks at zero spend; spend +18% vs the weeks before), and the small revenue dip was normal noise (50% of 141 windows of comparable periods show a gap as big). Before seeing the test, our model put Channel3 at 1.23 (90%: 0.46-2.18) in those regions and weeks, so even taken at face value the vendor's number doesn't contradict it. Counting the vendor's result at a quarter of its weight moves Channel3 from 1.08 to 1.05; believing it fully gives 0.97. **The decision does not depend on it.**

**Proposed test.** The channel we know least about is Channel4 (not knowing its true return costs about 88k a quarter), but no affordable test can measure it: even 20 regions for 12 weeks has only 41% power. Instead, switch **Channel3** off in **20 of 40 regions**, picked at random in size-matched pairs, for **12 weeks** (plus 2 weeks of follow-up). It has 83% power to detect a channel that pays for itself and measures its return to ±0.63. Expected cost: about 181k (revenue lost minus spend saved). **Condition:** spend records must show zero Channel3 spend in every test region and week. The test won't change next quarter's plan; it is the cheapest way to check the whole model independently.

## 5. How far to trust this

- **It predicted weeks it never saw.** We hid the last 13 weeks (2023-10-23 to 2024-01-15); the model's forecast of their total revenue was off by 0.02%, weekly error 3.8% vs 4.7% for simple guesses, and 12 of 13 weeks inside its range (the miss: Christmas week).
- **It works region by region on purpose.** On national totals the same method says Channel3 returns only 0.40 (vs 1.08), because adding regions up hides the evidence and makes our spend indistinguishable from competitor activity. Don't judge channels from national totals.
- **It accounts for what else moves sales**: competitor activity, public sentiment, promotions and season. Without competitor activity, ads appear to *lose* money, because we spend most when rivals are busiest.

## 6. What the model cannot tell you

- **Profit.** We have revenue, not margins. Apply your margin to every return above.
- **The exact shape of diminishing returns.** The data shows returns fall, not precisely how fast. We checked the plan under fast and slow assumptions; the direction holds, the sizes move.
- **Long-term brand effects.** It sees effects lasting up to about 8 weeks; anything slower is not counted.
- **Proof of cause.** It assumes regional spend wasn't aimed at regions already expected to sell more. Only a randomised test, like the one proposed, settles that.
- **Anything new.** New channels, new creative, price changes or holidays (no holiday effect is modelled).

Method: Bayesian marketing mix model (Google Meridian) by region, with carry-over and diminishing returns, calibrated with the vendor test at 25% weight. Full choices and data checks: DECISIONS.md, DATA_ISSUES.md.